

Course Description

“How do we transition from writing basic code to building robust, reproducible pipelines that actually ‘do science’?”

This course is the second half of the Introduction to R sequence. Building directly on R I, we shift our focus from R as a programming language to R as a data-analysis platform. We explore the tidyverse philosophy, master ggplot2 for publication-ready visualization, and work through the statistical toolkit used every day in biological research: hypothesis testing, goodness-of-fit tests, t-tests, ANOVA, correlation, and regression.

- R is a table-stakes language for data science and bioinformatics.
- The tidyverse makes complex data pipelines readable and maintainable.
- ggplot2 produces publication-quality figures with far less effort than most alternatives.
- R's statistical ecosystem is unmatched for the tests used in biological research

Course Learning Outcomes

By the end of this course, you will be able to:

- Compare the syntax and structure of base R with the principles of the tidyverse.
- Use dplyr to wrangle and clean messy biological datasets using a readable, pipeable grammar.
- Use ggplot2 to produce layered, publication-quality visualizations.
- Conduct and interpret standard parametric and nonparametric hypothesis tests.
- Apply ANOVA, post-hoc tests, correlation, and linear regression to biological data.
- Check statistical assumptions rigorously before choosing and running any test.

You will be comfortable using the tidyverse to wrangle/clean data, produce publication-quality ggplot2 figures, and conduct and interpret standard parametric and nonparametric statistical tests in R.

Pre-requisites:

- Completion of Introduction to R I (or demonstrated equivalent competency) is required.
- Biostatistics I (or equivalent) is strongly recommended. This course focuses on using R to apply statistical methods, not on teaching statistics from first principles. Brief refreshers are provided at the start of relevant modules, but prior familiarity is assumed.

Course Overview

Introduction to R I and Introduction to R II form a complete, two-course sequence. Together they take you from zero programming experience to conducting, visualizing, and statistically interpreting biological data analyses in a fully reproducible workflow.

Introduction to R I

"Learn R as a programming language"

- Base R syntax & data types
- Data structures: vectors, lists, data frames, matrices
- Control flow, loops, custom functions
- Data wrangling & cleaning
- Base R visualization & descriptive statistics

Introduction to R II

"Use R as a data-analysis platform"

- Tidyverse philosophy & dplyr
- ggplot2 for publication-quality visualization
- Hypothesis testing (Chi-squared, Fisher's exact, t-tests)
- ANOVA & nonparametric alternatives
- Correlation & linear regression

Undergraduate Credit Track

ABOUT

This section applies only to students enrolled for university credit through the University of Maine. Internal-only JAX participants (no credit hours) follow the standard track and do not submit the items listed here.

Credit Requirements

1 credit unit (approximately 45 hours total: 12.5 hours contact time + independent project work)

- Attend all five modules and actively participate in in-class exercises.
- Complete all five standard module assignments (same as internal students).
- Complete the five undergraduate credit questions (one per module) using the JAXMouse dataset.
- Submit each module's Project Milestone file by the date posted on Canvas.
- Your Module 5 compiled report forms the R I component of the semester project, continued in R II.

Shared Dataset: JAXMouse Gene Expression (Simulated)

All undergraduate credit questions across both courses use a single simulated RNA-seq-style dataset:

JAXMouse_GeneExpression_Simulated.csv.

Column	Description
SampleID	Unique sample identifier (e.g., Brain_CTL_Rep1)
Tissue	Factor: Brain, Liver, Muscle
Treatment	Factor: Control, Treated
Replicate	Integer: 1, 2, or 3
Sex	Factor: Male, Female
Age_weeks	Numeric: mouse age at sacrifice (weeks)
Gene	Character: gene symbol (e.g., Actb, Gapdh, Brca1)
Expression	Numeric: log2 CPM expression value
QC_pass	Logical: TRUE / FALSE quality control flag
BatchID	Factor: Batch1, Batch2 (used to illustrate Simpson's Paradox)

Project Milestones: One Per Module

Module	Milestone Deliverable	Due
Module 1	JAXMouse_clean.csv (factors converted, columns verified)	<i>See GitHub</i>
Module 2	JAXMouse_gene_summary.csv (mean, SD, outlier flag per gene)	<i>See GitHub</i>
Module 3	JAXMouse_QCpass.csv (final clean, QC-filtered dataset)	<i>See GitHub</i>
Module 4	JAXMouse_baseR_plots.pdf (base R visualization suite)	<i>See GitHub</i>
Module 5	JAXMouse_RI_Analysis report (compiled Quarto HTML)	<i>See GitHub</i>

Schedule & Content

Module	Topic & Content	Resources
MODULE 0 — Setup & Orientation <i>Q: Are we ready to pick up where Intro to R I left off?</i> <i>A: Verify that R/RStudio are installed and run correctly and refresh the core R I skills we will build upon throughout this course.</i>		
0	Learning Objectives - Confirm R and RStudio (Posit) are installed and functional. - Recall key R I concepts: data structures, slicing, tapply(), base R plots. - Review the course overview and confirm prerequisites are met. In-Class Activities Self-check: run a short R I review script provided on Canvas before Day 1.	Installation JAX Software Catalogue (internal) General R Reference R for Data Science (Hadley Wickham) Intro to R (online book) R I refresher R for Data Science Stats Flowchart
MODULE 1 — Reviewing the Foundations: from R I to R II <i>Q: How do we leverage our foundational coding skills to efficiently interrogate a new biological dataset?</i> <i>A: We consolidate R I competencies, like data import, slicing, tapply(), base R plots, before layering the tidyverse on top. Everything you learned in R I is still valid; the tidyverse simply extends it.</i>		
1AB	Learning Objectives - Reload and inspect datasets using the full suite of base R tools (summary(), str(), head(), dim()). - Apply slicing with conditionals, subset(), and the *apply family. - Produce scatterplots, histograms, and mosaic plots in base R. - Articulate what you find clunky or repetitive about base R syntax, this motivates the tidyverse. - Understand the R II workflow: Tidy → Transform → Visualize → Model → Interpret In-Class Activities - Produce graphs with manipulated datasets in each of the above categories. - Side-by-side comparison: base R approach vs. a preview of the tidyverse approach.	Resources: ggplot2 cheat sheet Data Visualization with ggplot2 Assignment for Certificate of Completion: See GitHub Course Site.
MODULE 2 — Introduction to the Tidyverse: Tidy Data & dplyr & ggplot2 <i>Q: What exactly makes data "tidy," and why does it transform the way we write analytical code? How do we move beyond charts to construct layered, publication-ready visualizations that expose the truth in our data?</i> <i>A: We transition from the clunkiness of base R to the elegance of the tidyverse's grammar of data manipulation, using the dplyr package. We compare visualizations made in base R to the far more flexible and attractive results from ggplot2, understanding the "grammar of graphics" that makes ggplot2 composable.</i>		

2AB

Learning Objectives

- Define "tidy data" (one observation per row, one variable per column) and identify violations in real datasets.
- Work with tibbles and understand how they differ from base R data frames.
- Use the core dplyr verbs: filter(), select(), mutate(), arrange(), group_by(), summarize().
- Chain operations with the pipe operator (|> or %>%) for readable, sequential pipelines.
- Pass a dplyr result directly into ggplot2 using the pipe.
- Explain the three essential components of any ggplot2 plot: data, aesthetics (aes()), and geoms.
- Build plots layer by layer: geom_point(), geom_histogram(), geom_boxplot(), geom_bar().
- Modify aesthetics for color, fill, size, shape, and alpha.
- Add facets (facet_wrap(), facet_grid()) to show data across subgroups.
- Apply themes and custom labels for publication-ready output.
- Compare the same visualization produced in base R vs. ggplot2 and critique both.

In-Class Activities

- Manipulate and graph a built-in dataset using a dplyr + ggplot2 pipeline.
- Clean a messy dataset using tidyverse tools.
- Recreate base R Module 3 visualizations from R I using ggplot2 for a direct comparison.
- Explore a new dataset and produce a ggplot2 visualization suite.

Tidyverse guides

- [Tidy Data Tutor \(visual\)](#)
- [R4DS — Hadley Wickham](#)
- [GenZ dplyr explainer](#)

ggplot2 cheat sheet

- [Data Visualization with ggplot2](#)

Further reading

- [R4DS — Data visualization](#)

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 3 — Hypothesis Testing: Developing an Assumptions Pipeline

Q: How do we create a reproducible pipeline to ensure the assumptions of our powerful parametric statistical tests are met?

R: We establish a system of quick visualizations and built-in tests to test for normality.

3AB

Learning Objectives

- Apply the four-step hypothesis testing framework: state $H_0/H_a \rightarrow$ choose and run the test $\rightarrow$ compare p-value to $\alpha \rightarrow$ conclude.
- Distinguish Type I and Type II errors and understand statistical power.
- Build a binomial distribution and apply binom.test() and prop.test().
- Test normality using QQ plots and the Shapiro-Wilk test.
- Calculate and interpret ROC/AUC for diagnostic classifiers.

In-Class Activities

- Formally test a hypothesis on the Bumpus sparrow dataset (binomial test).
- Reproduce a ROC/AUC example using the pROC library.

Reference

- [Hypothesis testing \(Wikipedia\)](#)

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 4 — Hypothesis Testing: t-Tests & Survival Analysis

Q: How do we translate a biological question into a mathematically rigorous, code-based hypothesis test?

A: We establish a universal four-step framework for hypothesis testing and apply it to an escalating series of tests: from binomial proportions through goodness-of-fit, t-tests, and Kaplan-Meier survival curves.

4AB**Learning Objectives**

- Conduct one-sample, two-sample, and paired t-tests; identify the correct test for the experimental design.
- Apply nonparametric alternatives (Wilcoxon test, Welch's t-test) when normality assumptions are violated.
- Construct and interpret Kaplan-Meier survival curves.

In-Class Activities

- Produce a Kaplan-Meier curve for the built-in leukemia dataset.

Reference

- [Type I & II errors \(visual\)](#)
- [Power demo 1](#)
- [Power demo 2](#)

Assignment for Certificate of Completion: See GitHub Course Site.

MODULE 5 — ANOVA, Correlation & Linear Regression

Q: How can we reliably predict outcomes and map complex relationships between multiple variables while accounting for biological noise and outliers?

A: We extend hypothesis testing from two-group comparisons to multi-group ANOVA and continuous relationships via correlation and linear regression, always checking assumptions and investigating residuals.

5AB**Learning Objectives**

- Test homogeneity of variances with `bartlett.test()` before running ANOVA.
- Run a one-way ANOVA and apply `TukeyHSD()` for post-hoc pairwise comparisons.
- Run a Welch's Approximate ANOVA when groups are normally distributed, but variances are not equal.
- Apply the nonparametric alternative to ANOVA: `kruskal.test()`.
- Distinguish between correlation (measuring association) and regression (predicting values).
- Calculate Pearson's *r* and its nonparametric alternatives (Spearman, Kendall).
- Fit a simple linear regression, interpret coefficients, and calculate confidence intervals.
- Produce and interpret a residual plot to identify outliers and check model assumptions.

In-Class Activities

- Cuckoo egg ANOVA: test whether egg size differs across host species.
- Mammal brain-body size regression: log-transform, fit, interpret residuals, identify outliers.

Simulations

- [Visual ANOVA \(Wolfram\)](#)
- [ANOVA video](#)
- [Regression fit demo](#)
- [Correlation vs. Regression game](#)

Assignment for Certificate of Completion: See GitHub Course Site.

Further Learning & Enrichment Resources

The resources below supplement the course content. None are required; they are provided for students who want to go deeper.

Resource	URL / Notes
R for Data Science (R4DS)	https://r4ds.hadley.nz/
tidyverse documentation	https://www.tidyverse.org/
ggplot2 cheat sheet	https://posit.co/wp-content/uploads/2022/10/data-visualization-1.pdf
Tidy Data Tutor (visual dplyr)	https://tidydatatutor.com
Statistics flowchart	https://www.statsflowchart.co.uk
Visual ANOVA (Wolfram)	https://demonstrations.wolfram.com/VisualANOVA
Correlation vs. Regression game	http://shiny.stat.calpoly.edu/Corr_Reg_Game/